

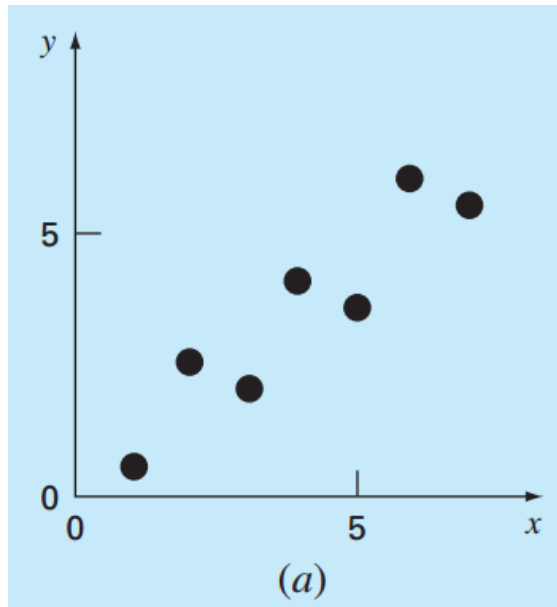
INTRODUCTION TO APPLIED NUMERICAL METHODS

Curve Fitting: Linear Regression

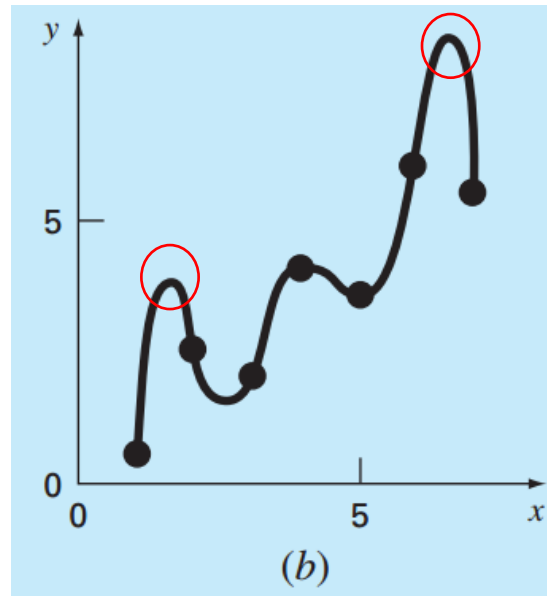
Lang Yuan, Associate Professor



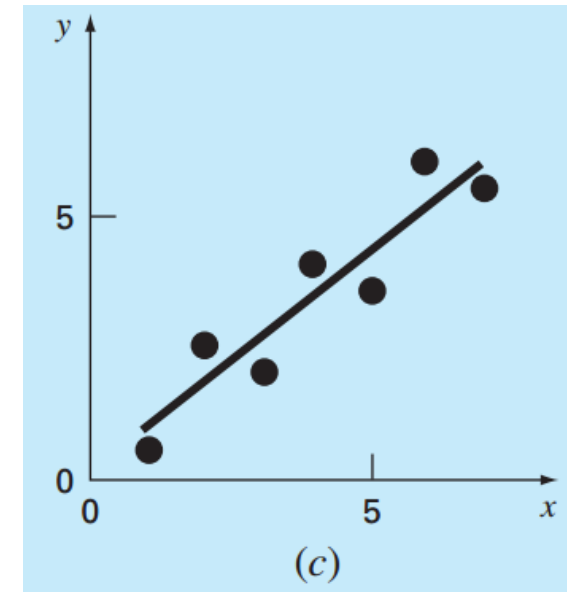
LEAST-SQUARES REGRESSION



Significant variability



Sixth-order Interpolating Polynomial



Linear Regression
(General Trend)

LINEAR REGRESSION

Linear Regression

- Fitting a straight line to a set of paired observations $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$

A Straight Line for Regression

$$y = a_0 + a_1x + e$$

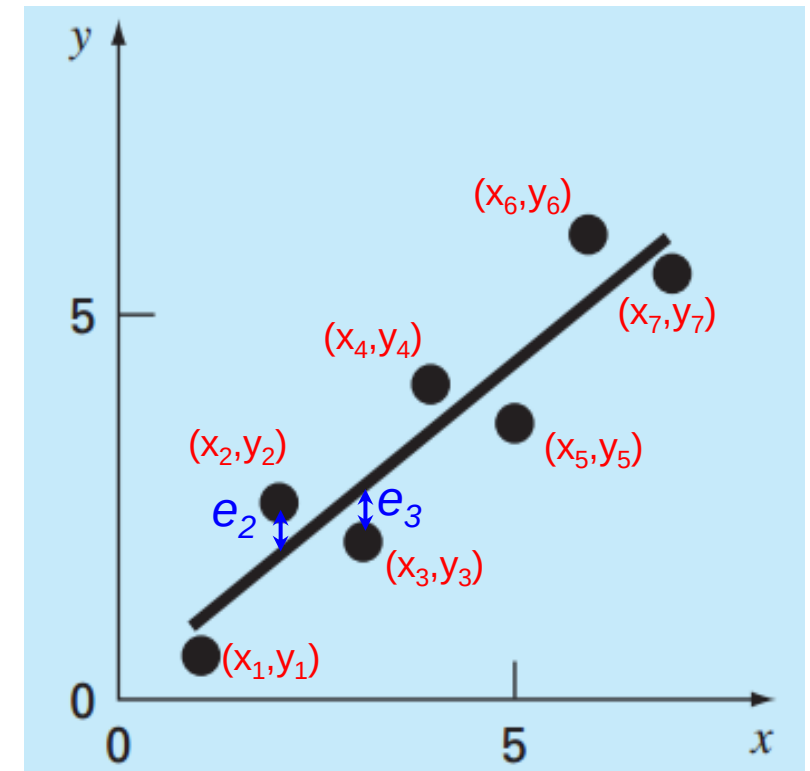
intercept

Slope

Error/residual

Residual

- Discrepancy between the true value of y and the predicted value predicted by the linear equation, $y_{model} = a_0 + a_1x$



CRITERIA FOR A “BEST” FIT

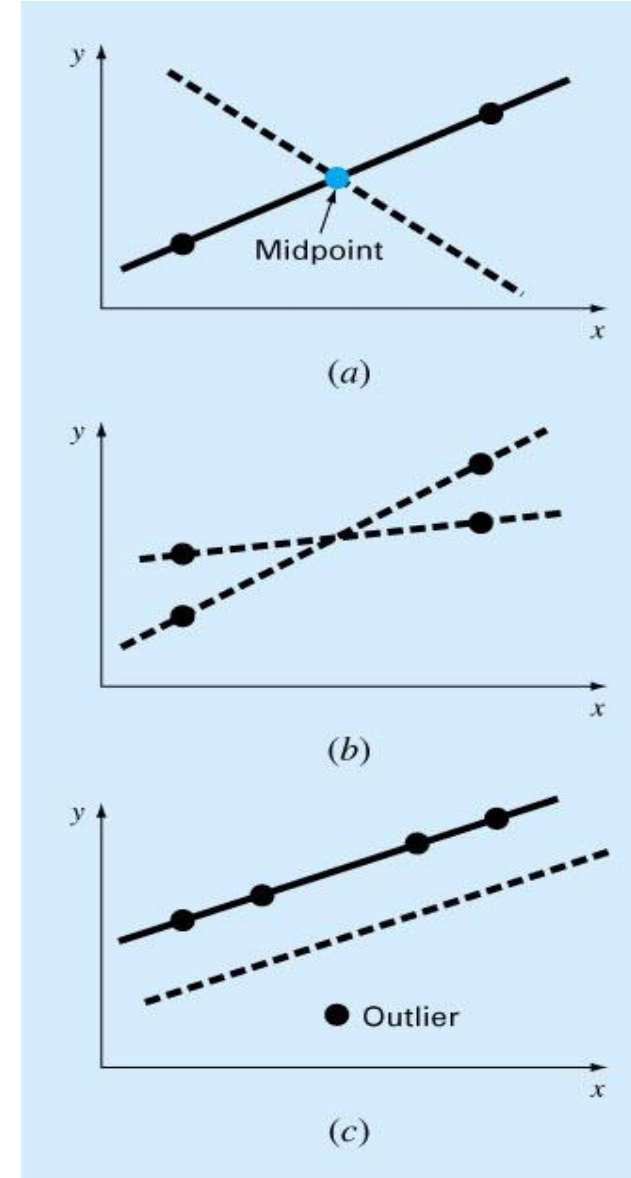
- Minimize the sum of the residual errors for all available data:

$$\sum_{i=1}^n e_i = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)$$

where n = total number of points.

- However, this is an inadequate criterion, so is the sum of the absolute values

$$\sum_{i=1}^n |e_i| = \sum_{i=1}^n |y_i - a_0 - a_1 x_i|$$



(a) minimize the sum of the residuals

(b) minimize the sum of the absolute values of the residuals

(c) minimize the maximum error of any individual point.

**engineering
and Computing**

CRITERION FOR FIT

Sum of the Squares of the Residual

- Measured $y_{i,\text{measured}}$ vs. predicted $y_{i,\text{model}} = a_0 + a_1 x_i$

$$S_r = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_{i,\text{measured}} - y_{i,\text{model}})^2 = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2$$

- Squares ensure no “cancelling” of error/residual while providing smooth derivative
- Yield a unique line for a given set of data

Goal: Identify the best combination of a_0 and a_1 yielding minimum “sum of the squares of the residual”

LEAST-SQUARES FIT OF A STRAIGHT LINE

Sum of the Squares of the Residual

- Measured $y_{i,\text{measured}}$ vs. predicted $y_{i,\text{model}} = a_0 + a_1 x_i$

$$S_r = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_{i,\text{measured}} - y_{i,\text{model}})^2 = \sum_{i=1}^n (y_i - a_0 - a_1 x_i)^2$$

$$\frac{\partial S_r}{\partial a_0} = -2 \sum (y_i - a_0 - a_1 x_i)$$

$$\begin{aligned} \frac{\partial S_r}{\partial a_0} &= \sum_{i=1}^n \frac{\partial (y_i - a_0 - a_1 x_i)^2}{\partial a_0} = \sum_{i=1}^n 2(y_i - a_0 - a_1 x_i) \frac{\partial (y_i - a_0 - a_1 x_i)}{\partial a_0} \\ &= 2 \sum_{i=1}^n (y_i - a_0 - a_1 x_i)(-1) = -2 \sum_{i=1}^n (y_i - a_0 - a_1 x_i) \end{aligned}$$

- Differentiate “each individual term” in “ Σ ” with respect to a_0

- Use the “chain rule”

- Differentiate “each individual term” in “ Σ ” with respect to a_1

- Use the “chain rule”

$$\frac{\partial S_r}{\partial a_1} = -2 \sum [(y_i - a_0 - a_1 x_i) x_i]$$

$$\begin{aligned} \frac{\partial S_r}{\partial a_1} &= \sum_{i=1}^n \frac{\partial (y_i - a_0 - a_1 x_i)^2}{\partial a_1} = \sum_{i=1}^n 2(y_i - a_0 - a_1 x_i) \frac{\partial (y_i - a_0 - a_1 x_i)}{\partial a_1} \\ &= 2 \sum_{i=1}^n (y_i - a_0 - a_1 x_i)(-x_i) = -2 \sum_{i=1}^n (y_i - a_0 - a_1 x_i) x_i \end{aligned}$$

LEAST-SQUARES FIT OF A STRAIGHT LINE

$$\frac{\partial S_r}{\partial a_0} = -2 \sum (y_i - a_0 - a_1 x_i) = 0$$

$$0 = \sum y_i - \sum a_0 - \sum a_1 x_i$$

Constant Constant

$$na_0 + \left(\sum x_i\right)a_1 = \sum y_i$$

$$\frac{\partial S_r}{\partial a_1} = -2 \sum [(y_i - a_0 - a_1 x_i)x_i] = 0$$

$$0 = \sum y_i x_i - \sum a_0 x_i - \sum a_1 x_i^2$$

$$\left(\sum x_i\right)a_0 + \left(\sum x_i^2\right)a_1 = \sum x_i y_i$$

LEAST-SQUARES FIT OF A STRAIGHT LINE

Two Equations and Two unknowns: a_0 and a_1

$$\begin{cases} na_0 + \left(\sum x_i\right)a_1 = \sum y_i \\ \left(\sum x_i\right)a_0 + \left(\sum x_i^2\right)a_1 = \sum x_i y_i \end{cases}$$

$\begin{matrix} \textcircled{1} \times (-\sum x_i) \\ + \\ \textcircled{2} \times n \end{matrix}$



$$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$



$$a_0 = \bar{y} - a_1 \bar{x}$$

Normal
Equations

Mean
of y

Mean
of x

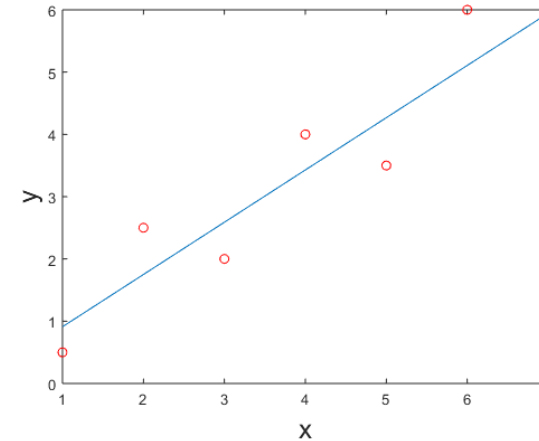
For Linear Regression: we need

$$\begin{cases} \sum_{i=1}^n x_i y_i \\ \sum_{i=1}^n x_i^2 \\ \sum_{i=1}^n x_i \quad \text{or} \quad \bar{x} \\ \sum_{i=1}^n y_i \quad \text{or} \quad \bar{y} \end{cases}$$

AN EXAMPLE

Linear regression of the x and y values in the first 2 columns in the Table

x_i^2	$x_i y_i$	x_i	y_i
1	0.5000	1	0.5
4	5.0000	2	2.5
9	6.0000	3	2.0
16	16.0000	4	4.0
25	17.5000	5	3.5
36	36.0000	6	6.0
49	38.5000	7	5.5
Σ		Σ	24.0



Solution:

First necessary quantities for linear regression will be computed:

$$n = 7 \quad \sum x_i y_i = 119.5 \quad \sum x_i^2 = 140$$

$$\sum x_i = 28 \quad \bar{x} = \frac{28}{7} = 4$$

$$\sum y_i = 24 \quad \bar{y} = \frac{24}{7} = 3.428571$$

$$a_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$$a_0 = \bar{y} - a_1 \bar{x}$$

$$a_1 = \frac{7(119.5) - 28(24)}{7(140) - (28)^2} = 0.8392857$$

$$a_0 = 3.428571 - 0.8392857(4) = 0.07142857$$

$$\Rightarrow y = 0.07142857 + 0.8392857x$$